Generative AI Overview

Generative AI refers to a class of artificial intelligence models that can create new content—such as text, images, audio, or code—rather than just analyzing existing data. These models learn patterns from large datasets and then generate outputs that resemble the training data while often adding novel variations.

Large Language Models (LLMs)

LLMs are a subset of generative AI focused on natural language. They are trained on massive corpora of text and can perform tasks like:

- Text completion

- Summarization

- Translation

- Question answering

Example: OpenAI's GPT■4 can take a prompt like "Explain quantum computing in simple terms" and produce a coherent, human■like paragraph.

Transformers

The transformer architecture, introduced in the 2017 paper *Attention Is All You Need*, revolutionized LLMs. Key components:

1. **Self■Attention:** Allows the model to weigh the relevance of each word in a sequence relative to others.

2. **Positional Encoding:** Adds information about word order.

3. **Feed■Forward Networks:** Process the attention outputs.

Transformers enable parallel processing of tokens, making training on huge datasets feasible.

Embeddings

Embeddings are dense vector representations of data (words, sentences, images) that capture semantic meaning. In language models, each token is mapped to an embedding vector before being processed by the transformer.

Example: The words "king" and "queen" have embeddings that are close in vector space, and the relationship vector (queen - king) is similar to (woman - man).

Retrieval■Augmented Generation (RAG)

RAG combines a generative model with an external knowledge base. The workflow:

1. **Retrieval:** A query is used to fetch relevant documents or passages from a vector store using embeddings.

2. **Augmentation:** Retrieved texts are appended to the prompt.

3. **Generation:** The LLM generates a response grounded in the retrieved information.

Example Use■Case: A customer■support bot that answers product questions by retrieving the latest FAQ entries and then generating a natural■language answer.

Putting It All Together

1. **Data Collection:** Gather large text corpora and domain■specific documents.

2. **Embedding Creation:** Encode documents into vectors for fast similarity search.

3. **Transformer Training:** Train or fine■tune a transformer■based LLM on the corpus.

4. **RAG Pipeline:** At inference, retrieve relevant embeddings, feed them to the LLM, and generate the final answer.

This combination yields powerful, up■to■date, and context■aware AI assistants.

Prepared by VERA, your personal desktop assistant.